# FELIX KLEIN

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## EDUCATION

**Technical University of Munich (TUM)** Munich
*B. Sc. Physik* 10/2025 - expected 10/2026

- Additional physics degree pursued to extend a mathematics-based analytical profile towards technical, computational and biophysical research
- Relevant focus: simulation, dynamics, applied physics and preparation for graduate study in Applied and Engineering Physics, ESPACE, Robotics, Cognition, Intelligence, or Mathematics in Data Science

**University of Mannheim** Mannheim
*B.Sc. Business Mathematics | Final grade: 2.3* 09/2020 - 09/2024

- Bachelor thesis: "Mathematical Brain Tumour Modelling Using the Example of Glioblastoma: Analysis of Cell Dynamics and Therapy Effects with Continuous-Time Markov Chains"
- Core areas: Markov chains, stochastic processes, dynamical systems, numerical methods, optimisation and scientific programming

**Western Sydney University** Sydney, Australia
*Exchange Semester | Mathematics and Computer Science | Grade: 1.7* 07/2023 - 02/2024

- Coursework in Machine Learning, Data Science and Mathematical Modeling
- Project work on convolutional neural networks for image classification and feature-map visualisation using PyTorch

**Rheingau Gymnasium** Geisenheim
*German Abitur | Final grade: 1.9* 07/2012 - 06/2020

- Awarded the DPG Physics Abitur Prize

## PRACTICAL AND PROFESSIONAL EXPERIENCE

**Deutsche Aircraft** Oberpfaffenhofen
*Working Student, Automation and Digitalisation* 02/2026 - present

- Modelling and optimisation of data flows between ERP and the SupplyOn supplier platform for the serial production ramp-up of the D328eco
- Identification of systematic bottlenecks in supplier communication and derivation of quantitative metrics for data quality, order confirmations and delivery-date reliability

**Technical University of Munich - Chair of Agricultural Mechatronics** Garching
*Student Research Assistant (HiWi), Collaborative Routing and Simulation* 02/2026 - present

- Literature review on collaborative multi-agent routing algorithms and Controlled Traffic Farming as a combinatorial optimisation problem
- Development of a Python simulation environment for evaluating route-allocation and consensus algorithms on realistic agricultural field graphs
- Implementation and performance evaluation of collaborative routing heuristics with respect to runtime, solution quality and scalability; reproducible documentation with Sphinx

**BMW Group – Research and Innovation Centre** Munich
*Intern, Engineering Operations & Data Analytics* 04/2025 - 09/2025

- Developed a forecasting pipeline for axle-assembly demand in prototype production using Python, time-series modelling and Power BI visualization

- Designed a discrete-event simulation model for warehouse utilisation at the Parsdorf supply centre, quantifying bottlenecks in inventory and disposal logistics
- Supported the NA0 VS0 8/26 prototype cubing process through CATIA V5 and VisView geometry preparation, tolerance analysis and coordination between engineering, quality and suppliers
- Documented quality deviations in PQM Next in German and English, creating a structured data basis for statistical analysis in prototype testing

**KombiRail Europe B.V.** Frankfurt

*Intern, Executive Office & Strategy* 01 - 31/03/2025

- Prepared quantitative analyses of operational and financial performance indicators for senior management
- Modelled cost-saving scenarios of approximately EUR 400k p.a. as decision support for an M&A-related process

**KPMG - Lighthouse** Saarbrücken

*Intern, AI & Data Science Consulting* 09/2024 - 02/2025

- Extended the Python backend of a simulation software with numerical analysis modules; vectorisation and caching reduced runtime of complex scenarios by approximately 25%
- Automated UI regression tests for an Angular frontend with Selenium and integrated them into a Git/Azure DevOps CI pipeline, reducing manual testing effort by around 50% per sprint
- Refactored and optimised R scripts for stochastic reserve forecasting, supporting quantitative risk assessments for client projects

**Cosuno Ventures GmbH** Berlin

*Working Student, Revenue Operations & Sales Analytics Analyst* 08/2021 - 09/2022

- Designed automated reporting workflows using HubSpot, Excel/VBA and data-cleaning routines, reducing KPI report preparation from two days to three hours
- Supported weekly data-driven management updates and operational sales controlling in a venture-backed construction-tech start-up

**Family Winery Martin Klein** Johannisberg

*Business Development & E-Commerce – Family Winery* 01/2015 - present

- Contributed across operations, sales and customer service in the family winery, providing long-term practical exposure to operations, hospitality and end-to-end customer fulfilment
- Built and maintained an e-commerce and digital-marketing channel using WooCommerce, SEO, web analytics and conversion-data evaluation

## ACADEMIC AND APPLIED PROJECTS

**Hackathons & Datathons** Hamburg & Mannheim

*Participant and Team Contributor* 2023 & 2024

- 2nd place, ChefTreff Hackathon 2024 with Lufthansa Industry Solutions and ZeroG: stochastic forecasting pipeline using XGBoost and Monte Carlo simulation to optimise onboard catering under demand uncertainty
- STADS Datathon 2023, EY Challenge: Python-based matching algorithm for detecting duplicate financial transactions using fuzzy hashing and feature engineering; fraud heatmap visualisation for case-based savings analysis

## ENGAGEMENT AND LEADERSHIP

**HORYZN Student Aerospace Initiative** Munich

*Aerostructures Team - Landing Gear* 01/05 - present

- Responsible for the development and optimisation of landing-gear concepts for autonomous aircraft prototypes

- Analysing landing stability, shock absorption, structural constraints and manufacturability; supporting CAD prototyping and interdisciplinary aerospace design work

**Volunteer Teacher – Indian High School, Jaipur** Jaipur, India
*Afternoon Programme, approximately five days per week* 18/08 - 31/09/2025

- Designed and taught a practical "English for Job Interviews" course for students in grades 10-12, including structured speaking practice and simulated interviews
- Developed a "Digital Literacy and AI" module in which students created CVs, used generative AI for guided research and built small presentation-based or programming-oriented mini projects

**TG Rüdesheim „JuSki"** Mühlbach am Hochkönig, Austria
*Snowboard Instructor* 01/2024 & 01/2025

- Instructor for approximately 50 teenagers aged 14-17 as part of a 12-person supervision and coaching team

**STADS – Student Association for Data Analytics & Statistics** Mannheim
*Member* 09/2022 - 09/2025

- Participated in more than 20 workshops and guest lectures on data science, statistics and machine learning with companies including SAP and Mercedes-Benz

**Student Council Mathematics - University of Mannheim** Mannheim
*Quality Assurance and Budget Allocation* 09/2022 - 09/2023

- Contributed to the allocation of an annual student-body budget of approximately EUR 500k and supported a transparent, department-based prioritisation process

## SKILLS

Languages: German (native) · English (C2) · Spanish (B1) · Japanese (A2)
Programming and data science: Python (NumPy, Pandas, Scikit-Learn, PyTorch) · Julia · R · MATLAB · basic C++
Mathematical methods: Markov chains · stochastic processes · dynamical systems · numerical optimisation · Monte Carlo simulation · machine learning
Tools: Git · LaTeX · Sphinx · Power BI · Azure DevOps CI/CD · CATIA V5 · VisView · HubSpot · Salesforce
Interests: Applied physics · automation · endurance running · snowboarding · surfing and hands-on technical prototyping